STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 10: Thermal Properties of Matter

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. What is temperature’s role at equilibrium?
2. Convert 27 deg C to kelvin.
3. Specific heat capacity?
4. Heat for temperature change without phase change?
5. Latent heat relation?
6. Heat transfer through vacuum?
7. Ideal gas equation?
8. Newton’s law of cooling?

Section B 2-mark questions - 6 x 2 = 12 marks

9. kg warmed 5 K, c=400: Q?
10. alpha=1.2 x 10-,L=2m,DeltaT=50K: extension?
11. 100 deg C in kelvin?
12. Heat capacity of .5 kg with c=800?
13. Heat to melt .2 kg ice, L=3.34 x 10?
14. Why is temperature constant during boiling?

Section C 3-mark questions - 4 x 3 = 12 marks

15. .5 kg water c=4200 heated 10 deg C: heat?
16. Rod L=2m,alpha=1.5 x 10-,DeltaT=40K: expansion?
17. Mix .2 kg at 80 deg C and .3 kg at 20 deg C, equal c, no heat loss: final T?
18. Gas at fixed volume warms 300K to 450K; initial p=100kPa. Final p?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Describe three heat-transfer modes.
20. Melt .2 kg ice at 0 deg C then warm water by 20K; L=3.34 x 10,c=4200. Total Q?